

Practice1_Barraza_Torres_U2

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2025-06-19

Actividad 1: Venn-Euler diagram, data concept inside and outside of groups, etc.

In probabilities, the concept of groups encapsulates the way that 2 conditions shares the same parts of an established spectrum. For this, we need to comprehend the way data inside 2 (or more) groups works.

Importation on specific packages.

In this part, we need some packs about the way we can comprehend parts, being this the Venn-Euler diagram and the spectrum (also known as the universe).

```
#Importaciones
library(VennDiagram)

## Cargando paquete requerido: grid

## Cargando paquete requerido: futile.logger

library(tidyverse)

## — Attaching core tidyverse packages —————
tidyverse 2.0.0 —
## ✓ dplyr      1.1.4      ✓ readr      2.1.5
## ✓ forcats    1.0.0      ✓ stringr    1.5.1
## ✓ ggplot2     3.5.2      ✓ tibble     3.3.0
## ✓ lubridate  1.9.4      ✓ tidyr      1.3.1
## ✓ purrr      1.0.4

## — Conflicts —————
tidyverse_conflicts() —
## ✗ dplyr::filter() masks stats::filter()
## ✗ dplyr::lag()     masks stats::lag()
## ⓘ Use the conflicted package (<http://conflicted.r-lib.org/>) to
force all conflicts to become errors
```

Universe and conditions (sets)

We establish here the universe where we have the things (in this case, numbers from 1 to 20) and 3 sets (A, B, C) with numbers on it

```
###Universe (rango de datos)
```

```
universe <- 1:20
```

```
###Conjuntos
```

```
A <- c(8, 10, 18, 17, 11, 12, 16)
```

```
B <- c(2, 6, 6, 17, 11)
```

```
C <- c(18, 8, 9, 3)
```

Unions

Unions are the things some sets share between them without repeating any other data inside them

```
###Unión (U)
```

```
union_ab <- union(A, B)
```

```
print(union_ab)
```

```
## [1] 8 10 18 17 11 12 16 2 6
```

```
##ACT 1
```

```
#A U C
```

```
union_ac <- union(A, C)
```

```
print(union_ac)
```

```
## [1] 8 10 18 17 11 12 16 9 3
```

```
#B U C
```

```
union_bc <- union(B, C)
```

```
print(union_bc)
```

```
## [1] 2 6 17 11 18 8 9 3
```

```
#A U B U C
```

```
union_abc <- union(A, union(B, C))
```

```
cat(union_abc)
```

```
## 8 10 18 17 11 12 16 2 6 9 3
```

Intersection

Intersections are the data shared between some sets. This only applies when the data is in both sets at the same time.

```
###Intersección (n)
```

```
interseccion_ab <- intersect(A, B)
```

```
print(interseccion_ab)
```

```
## [1] 17 11
```

```

##ACT 2
#A ∩ C
interseccion_ac <- intersect(A, C)
print(interseccion_ac)

## [1] 8 18

#B ∩ C
interseccion_bc <- intersect(B, C)
print(interseccion_bc)

## numeric(0)

#A ∩ B ∩ C
interseccion_abc <- intersect(A, intersect(B, C))
cat(interseccion_abc)

```

Difference

Differences are the data that is in one set that is NOT in the other set

```

###Diferencia (-)
diff_ab <- setdiff(A, B)
print(diff_ab)

## [1] 8 10 18 12 16

diff_ba <- setdiff(B, A)
print(diff_ba)

## [1] 2 6

##ACT 3
#A - C
diff_ac <- setdiff(A, C)
print(diff_ac)

## [1] 10 17 11 12 16

#C - A
diff_ca <- setdiff(C, A)
print(diff_ca)

## [1] 9 3

#B - C
diff_bc <- setdiff(B, C)
print(diff_bc)

## [1] 2 6 17 11

```

```

#C - B
diff_cb <- setdiff(C, B)
print(diff_cb)

## [1] 18  8  9  3

#B - (A U C)
diff_bonac <- setdiff(B, union(A, C))
cat(diff_bonac)

## 2 6

```

Complement

Complements are the opposite of sets, it represents the data that is NOT in one set, it includes universe as a thing.

```

###Complementos (')
complement_a <- setdiff(universe, A)
print(complement_a)

## [1]  1  2  3  4  5  6  7  9 13 14 15 19 20

##Act 4
#B'
complement_b <- setdiff(universe, B)
print(complement_b)

## [1]  1  3  4  5  7  8  9 10 12 13 14 15 16 18 19 20

#C'
complement_c <- setdiff(universe, C)
print(complement_c)

## [1]  1  2  4  5  6  7 10 11 12 13 14 15 16 17 19 20

#(A U B U C)'
complement_abc <- setdiff(universe, union(A, union(B, C)))
cat(complement_abc)

## 1 4 5 7 13 14 15 19 20

```

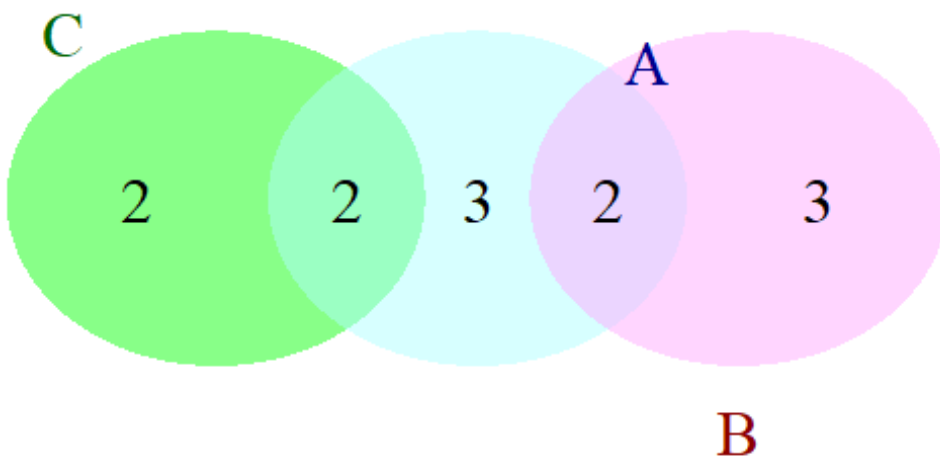
What is a complement?

The complement of an event A is defined as all outcomes in the sample space that are not part of A

Venn-Euler diagram

The venn-euler diagram is a type of representation of data inside a set in a universe, with the representation of data between them that they share (as a point, on my example i didn't had any data shared between all 3 of them, just an NRG coincidence)

```
venn.plot <- draw.triple.venn(  
  area1 = length(A),  
  area2 = length(B),  
  area3 = length(C),  
  n12 = length(interseccion_ab),  
  n23 = length(interseccion_bc),  
  n13 = length(interseccion_ac),  
  n123 = length(interseccion_abc),  
  category = c("A","B","C"),  
  fill = c("#ffffff", "#ffaaff", "#11ff11"),  
  lty = "blank",  
  cex = 2,  
  cat.cex = 2,  
  cat.col = c("darkblue", "darkred", "darkgreen")  
)
```



##Lo intenté xd

Conclusion

Data in probability must be shared between data properly because of the way data works, it is crucial to know more about this and suggest better chances